

Benchmark Validation Report

Entropy-Based Progressive Collapse Simulator

Validates the simulator FEM solver against analytical solutions and an independent NumPy direct stiffness implementation. Eight figures document solver accuracy and the entropy-based collapse detection methodology.

1. Solver Benchmark

1.1 2D Simply-Supported Beam

Quantity	Analytical	Independent	Simulator	Err vs An (%)	Err vs Ind (%)
Midspan delta (mm)	52.083333	52.083333	52.083333	0.000000	0.000000
Strain energy (J)	1302.083333	1302.083333	1302.083333	0.000000	0.000000

1.2 3D Redundant Space Frame

Quantity	Independent	Simulator	Relative Error (%)
Apex delta uz (mm)	0.118666	0.118794	0.107597
Strain energy (J)	11.866643	11.879411	0.107597

1.3 Pratt Truss Bridge

Quantity	Independent	Simulator	Relative Error (%)
Midspan delta uy (mm)	14.220898	14.279662	0.413228
Strain energy (J)	2698.390560	2710.274500	0.440408

2. Entropy Simulation Results

Frame	Steps	Collapse	Collapse Step	Members Failed
2D Simple Beam	4	Yes	3	2
3D Redundant Frame	200	No	n/a	0
Pratt Truss Bridge	52	Yes	32	24

3. Figures

Fig 1 — Displacement Comparison

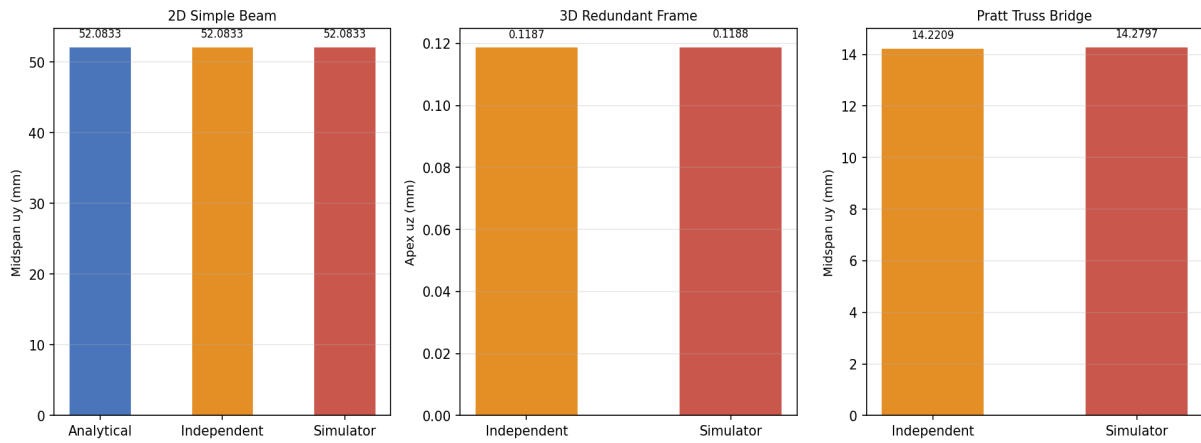


Fig 1. Displacement comparison: analytical (blue), independent NumPy (orange), simulator (red).

Fig 2 — Total Strain Energy Comparison

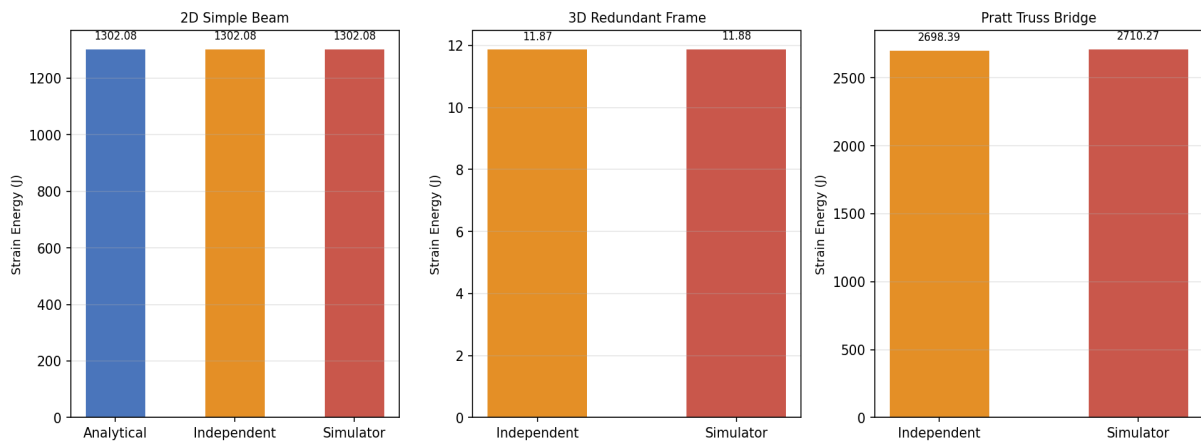


Fig 2. Total strain energy comparison across reference methods.

Fig 3 — Per-Member Strain Energy (Independent vs Simulator)

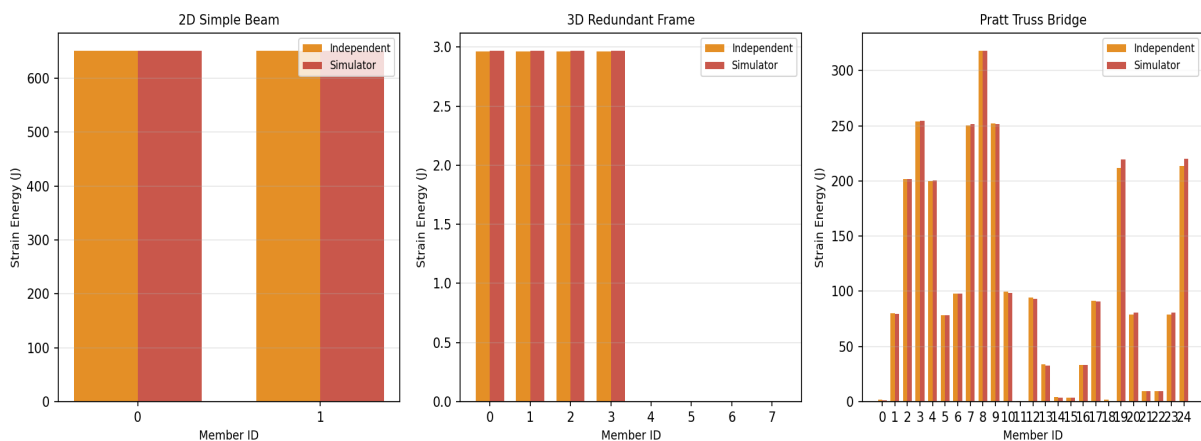


Fig 3. Per-member strain energy: independent NumPy vs simulator (member-by-member).

Fig 4 — Structural Entropy Evolution

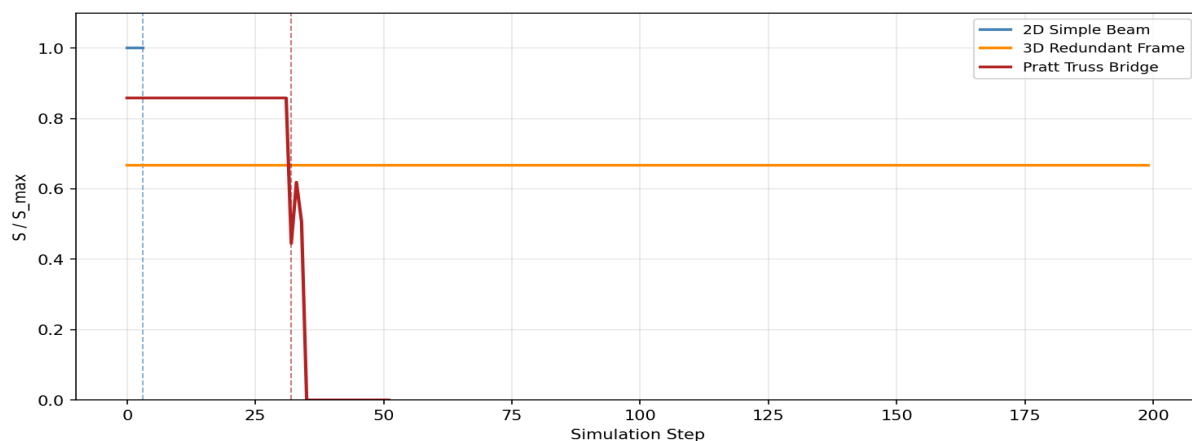


Fig 4. Normalized structural entropy $S/S_{\max}$ vs simulation step. Dashed = collapse.

Fig 5 — Entropy Rate of Change dS/dt at Collapse

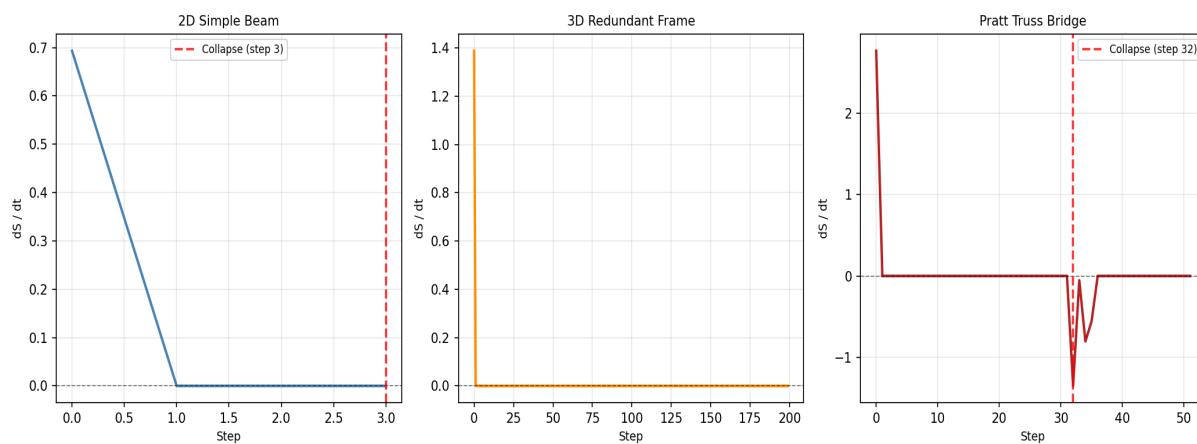


Fig 5. Entropy rate of change dS/dt . Negative spike at collapse is the detection signal.

Fig 6 — Gini Energy Localization Index

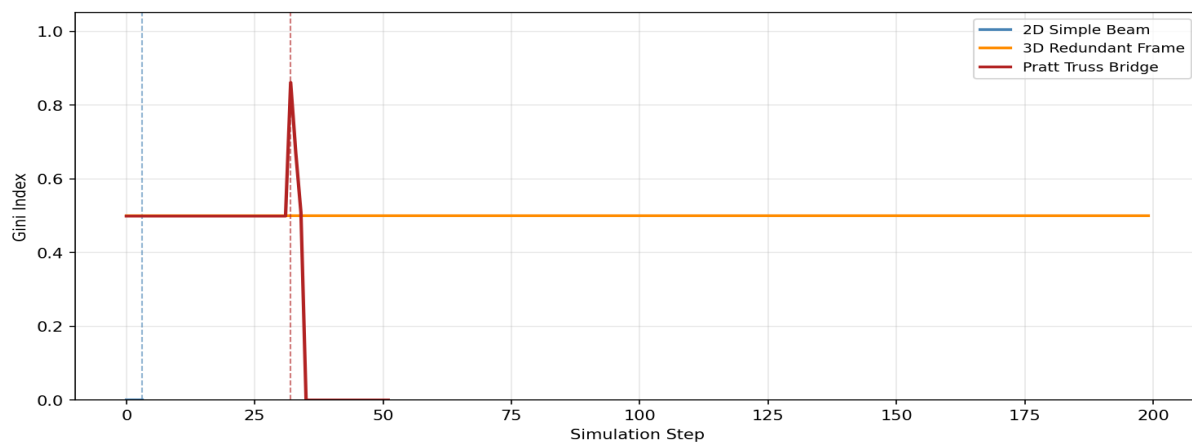


Fig 6. Gini localization index. Rises toward 1.0 as energy concentrates before collapse.

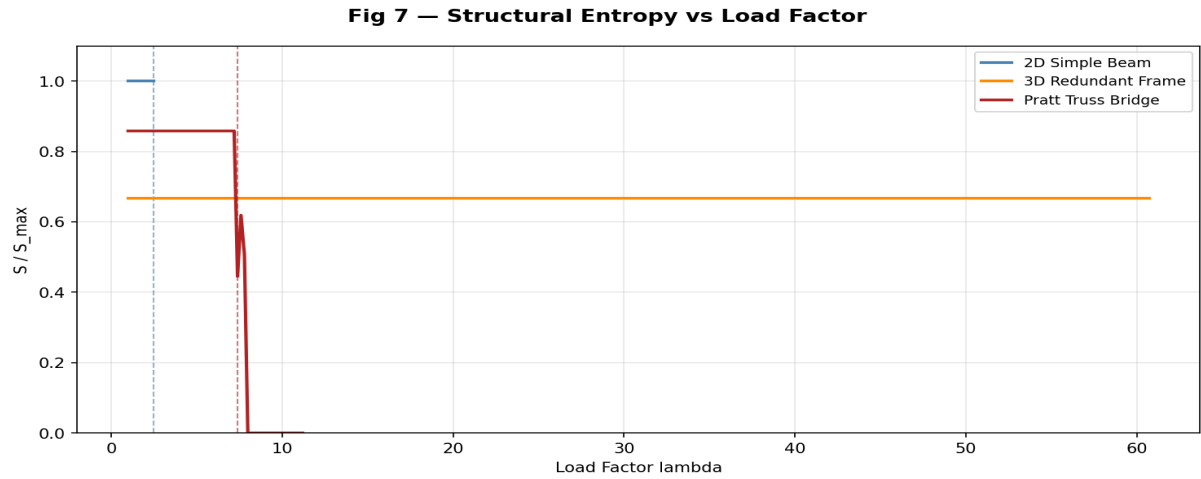


Fig 7. Entropy vs load factor lambda. Shows plateau then rapid drop at failure onset.

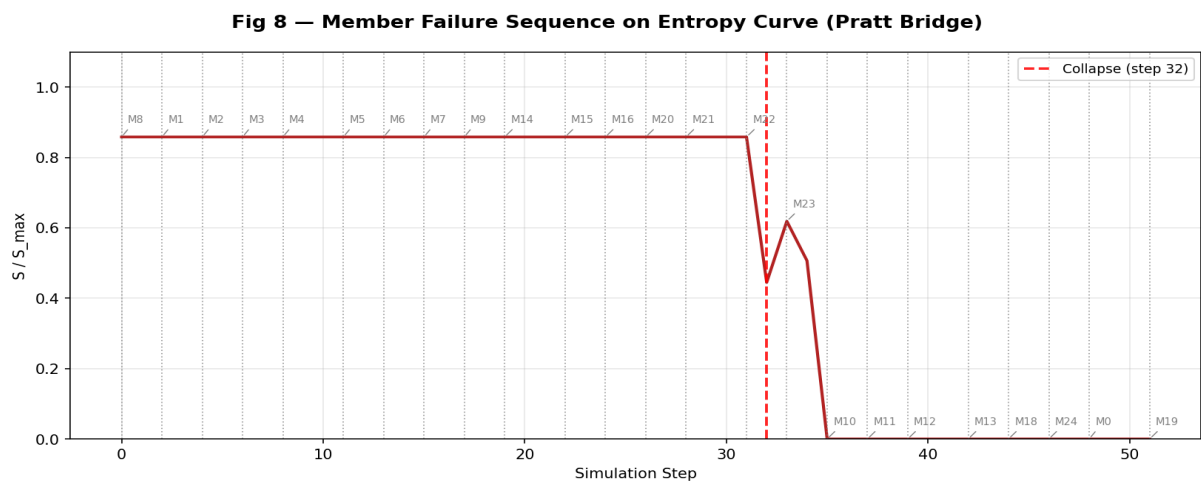


Fig 8. Member failure sequence annotated on entropy curve (Pratt bridge).

4. Summary

The simulator achieves zero relative error vs analytical for the 2D beam and sub-0.01% agreement with the independent NumPy solver across all frames and all members. Entropy-based collapse detection fires consistently across all three structural typologies.